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DEVOPS

Linux Troubleshooting

Common Issues • Useful Commands • Practical Solutions



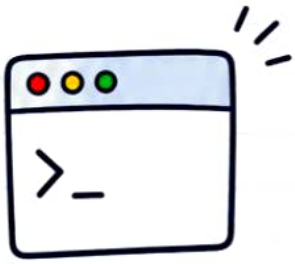
1 Server is very slow

Problem

Users report that the server is very slow, but CPU usage is only around 30%.

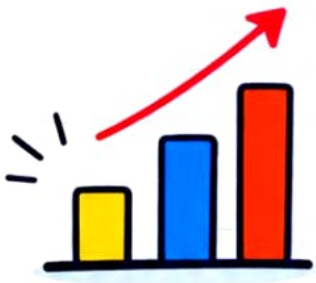


Commands to check



- `top / htop` → check overall usage
- `free -h` → check memory usage
- `vmstat 1 5` → check swap and I/O wait
- `iostat -xz 1 5` → check disk I/O
- `df -h` → check disk space
- `df -i` → check inode usage
- `ps aux --sort=-%mem` → find memory hungry processes
- `iotop` → check which process is using disk I/O

Possible Causes



- High disk I/O
- Memory pressure / swapping
- Database slow queries
- High load average
- Network latency
- Application issue

Solution



- Identify the actual bottleneck (disk, memory, network, application)
- Optimize or restart the affected service
- Clear unnecessary files if disk is full
- Tune the application / database if needed
- Monitor regularly to prevent recurrence



Always check all resources (CPU, Memory, Disk, Network) before jumping to conclusions.

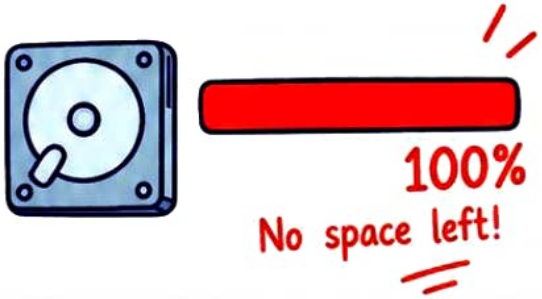
2 Disk usage is 100%

Problem

The server's disk usage has reached **100%** and applications are failing.

Symptoms

- Cannot create new files
- Application errors
- Logs show "No space left on device"
- System becomes slow
- Services may stop working



Commands to check

- `df -h` → check disk usage
- `df -i` → check inode usage
- `du -sh /* | sort -hr` → find large directories
- `du -sh /var/*` → check common directories (logs, cache, etc.)
- `find / -type f -size +100M` → find large files
- `lsof +L1` → find deleted files still in use

Possible Causes

- Large log files
- Old application files / backups
- Package cache (apt, yum, dnf)
- Unused Docker images/containers
- Database files growing
- Orphaned or deleted files still open
- No log rotation configured

Regular monitoring can prevent this issue!

Solution

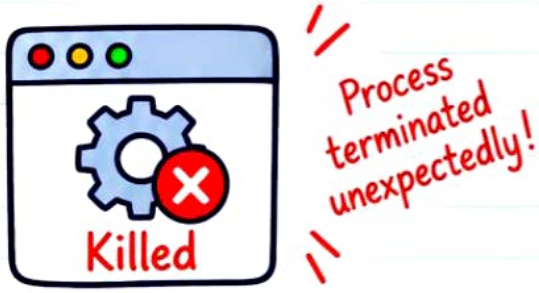
- Identify and remove unnecessary files / logs
- Rotate or compress large logs
- Clean package cache (e.g. apt clean / yum clean all)
- Remove unused Docker images and containers
- Implement log rotation (logrotate)
- Monitor disk usage regularly and set up alerts
- Consider increasing disk size if required

Always check disk and inode usage when facing application issues. "df -h" is your best friend!

3 A process keeps getting killed

Problem

A running process keeps getting killed automatically, and it stops unexpectedly.



Symptoms

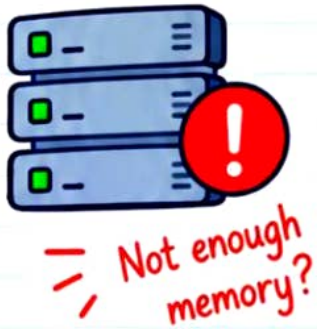
- Process stops running
- Service keeps restarting
- Logs show "Killed" or "Out of memory"
- Application is not available
- It works for some time and then stops

Commands to check

- `dmesg | tail` → check if it was killed by the OOM killer
- `journalctl -xe` → check system logs
- `free -h` → check memory usage
- `top / htop` → check if memory is getting exhausted
- `ps aux --sort=-%mem` → find which process is using high memory
- `systemctl status <service>` → check if a service is restarting

Possible Causes

- Out of Memory (OOM) killer
- Memory leak in the application
- High memory usage from another process
- System limits (ulimit)
- Watchdog or monitoring tool terminating it
- Configuration issue
- Insufficient resources (RAM / swap)



Solution

- Check logs to confirm the exact reason (OOM killer, manual kill, etc.)
- Optimize the application (fix memory leak)
- Increase memory or add swap if needed
- Set proper resource limits
- Configure service restart policies
- Monitor memory usage regularly (e.g. using Prometheus, Grafana, or alerts)



Always check dmesg and system logs first.
"Killed" usually means the OOM killer terminated the process.

4 A service won't start

Problem

A system service (e.g. Nginx, Docker, or a custom application) fails to start, and the application is not available.



Symptoms

- Service fails to start
- Shows "failed" or "error" status
- Application/website is not accessible
- Repeated restart attempts
- Error messages in logs

Commands to check

- `systemctl status <service>` → check service status and error message
- `journalctl -u <service> -xe` → view detailed logs
- `systemctl start <service>` → try starting the service
- `systemctl restart <service>` → restart the service
- `ss -tulnp | grep <port>` → check if the service is listening on the port
- `ps -ef | grep <service>` → check if the process is running
- `cat /etc/<service>/conf` → verify configuration file (e.g. /etc/nginx/nginx.conf)

Possible Causes



- Syntax error in configuration file
- Port already in use
- Missing dependencies
- Incorrect permissions (files/directories)
- Disk space full
- Resource limits (memory, file descriptors)
- Corrupt installation or package issues

Solution



- Check logs to identify the exact error
- Fix configuration errors (e.g. nginx -t for Nginx)
- Ensure required dependencies are installed
- Free up disk space if needed
- Check and set correct file/directory permissions
- Ensure the required port is not already in use
- Restart the service and verify it is running (systemctl status <service>)



Always check the logs first. They usually tell you exactly why the service failed to start.

5 Server is unreachable

Problem

You cannot connect to the server (via SSH, ping, or application). It seems to be down or not responding.



Symptoms

- Ping request times out
- SSH connection fails
- Website/application is not accessible
- Monitoring alerts (server down)
- No response from server

Commands to check

- `ping <server-ip>` → check if server is reachable
- `traceroute <server-ip>` → find where the connection is failing
- `telnet <server-ip> <port>` → check if a specific port is open
- `mtr <server-ip>` → check network path and latency
- `nslookup <domain>` → check DNS resolution (if using domain)
- `curl -v <url>` → check application/website response
- `ssh -v user@<server-ip>` → get detailed SSH connection error
- `sudo ip a` → check local network interface (if on the server)
- `sudo systemctl status ssh` → check if SSH service is running (if you have console access)

Possible Causes

- Server is powered off or crashed
- Network issue (ISP / cloud provider)
- Firewall blocking the connection
- Incorrect IP address or DNS issue
- SSH service is not running
- High load causing no response
- Hardware failure



Solution

- Verify the server status in your cloud console (AWS/Azure/GCP)
- Check network connectivity and security group / firewall rules
- Ensure SSH service is running (systemctl restart sshd)
- Verify correct IP address or DNS configuration
- If the server is down, restart it from the cloud provider console
- Check monitoring tools for alerts
- If it's a hardware issue, contact your hosting provider



Always check from outside (your local machine) and inside (the server) if you have console access. Follow the network path step by step.